

APPENDIX A
CENTRALIZED IHPS DISPATCH MODEL

Nomenclature

Indices and Sets:

$\mathcal{I}^{\text{CHP}}$	Index set of CHP units.
$\mathcal{I}^{\text{CU}}$	Index set of conventional non-CHP units.
$\mathcal{I}^{\text{DP}}$	Index set of DHN pipelines.
$\mathcal{I}^{\text{HL}}$	Index set of heat loads.
$\mathcal{I}^{\text{WP}}$	Index set of water pumps.
$\mathcal{I}^{\text{tl}}$	Index set of transmission lines in the EPS.
$\mathcal{I}^{\text{wind}}$	Index set of wind farms.
$\mathcal{K}$	Index set of scheduling intervals.
$\mathcal{N}^{\text{bus}}$	Index set of buses in the EPS.
$\mathcal{N}^{\text{nd}}$	Index set of nodes in the DHN.
$\text{Nd}_l^{\text{in}}/\text{Nd}_l^{\text{out}}$	Indices of the inlet/outlet nodes of pipeline l .
Nd_j^{WP}	Index of the DHN node connected to water pump j .
$\mathcal{S}_i^{\text{PS+}}/\mathcal{S}_i^{\text{PS-}}$	Index sets of supply pipelines leaving/entering DHN node i .
$\mathcal{S}_i^{\text{PR+}}/\mathcal{S}_i^{\text{PR-}}$	Index sets of return pipelines leaving/entering DHN node i .
$\mathcal{S}_i^{\text{HS}}/\mathcal{S}_i^{\text{HL}}$	Index sets of heat sources/loads connected to DHN node i .
$\mathcal{S}_b^{\text{CU}}/\mathcal{S}_b^{\text{CHP}}$	Index sets of non-CHP/CHP units connected to EPS bus b .
$\mathcal{S}_b^{\text{wind}}$	Index set of wind farms connected to EPS bus b .
$\mathcal{S}_b^{\text{WP}}$	Index set of water pumps connected to EPS bus b .

Parameters and Constants:

$a_{m,i}$	Cost coefficient of CHP unit i .
$c_{m,i}$	Cost coefficient of non-CHP unit i .
$\mathbf{D}_{n_t}^{\text{B}}$	Temporal derivative matrix of the Bernstein basis.
F_l	Transmission limit of line l .
$\mathbf{LD}_{b,k}^{\text{B}}$	Electric-load coefficient vector at bus b in interval k .
P_g^ζ/Q_g^ζ	Electric/heat output at extreme point ζ of the feasible operating region of CHP unit g .
$\underline{P}_i/\overline{P}_i$	Minimum/maximum power output of unit i .
$\overline{P}_{i,k}^{\text{w}}$	Available wind power of wind farm i in interval k .
$\mathbf{Q}_{d,k}^{\text{B}}$	Heat-demand coefficient vector of heat load d in interval k .
$r_i^{\text{up}}/r_i^{\text{dn}}$	Upward/downward ramping limit of unit i .
$SR^{\text{up}}/SR^{\text{dn}}$	System-wide upward/downward spinning-reserve requirement.

$SF_{l,b}$	Shift factor from bus b to transmission line l .
NK_g	Number of extreme points defining the feasible operating region of CHP unit g .
μ_l	Pressure-loss coefficient of pipeline l .
σ_i	Wind-curtailment penalty coefficient of wind farm i .
ρ, c, η_j	Water density, specific heat capacity, and efficiency of water pump j , respectively.
$\Delta t, n_t$	Scheduling-interval length and temporal Bernstein degree.

Variables:

$\mathbf{m}_{l,k}^{\text{S,B}}/\mathbf{m}_{l,k}^{\text{R,B}}$	Mass-flow coefficient vectors of pipeline l in the supply/return network.
$\mathbf{m}_{g,k}^{\text{G,B}}/\mathbf{m}_{d,k}^{\text{D,B}}$	Mass-flow coefficient vectors of heat source g /heat load d .
$\mathbf{pr}_{i,k}^{\text{S,B}}/\mathbf{pr}_{i,k}^{\text{R,B}}$	Supply/return pressure-head coefficient vectors at DHN node i .
$\mathbf{P}_{i,k}^{\text{B}}/\mathbf{P}_{i,k}^{\text{w,B}}$	Power-output coefficient vectors of generation unit i /wind farm i .
$\mathbf{P}_{j,k}^{\text{PUMP,B}}$	Power-consumption coefficient vector of water pump j .
$\mathbf{Q}_{g,k}^{\text{B}}$	Heat-output coefficient vector of heat source g .
$\mathbf{ru}_{i,k}^{\text{B}}/\mathbf{rd}_{i,k}^{\text{B}}$	Upward/downward spinning-reserve coefficient vectors of unit i .
$\mathbf{T}_{l,k}^{\text{OS,B}}/\mathbf{T}_{l,k}^{\text{OR,B}}$	Outlet-temperature coefficient vectors of pipeline l in the supply/return network.
$\mathbf{T}_{l,k}^{\text{IS,B}}/\mathbf{T}_{l,k}^{\text{IR,B}}$	Inlet-temperature coefficient vectors of pipeline l in the supply/return network.
$\mathbf{T}_{i,k}^{\text{NS,B}}/\mathbf{T}_{i,k}^{\text{NR,B}}$	Supply/return nodal-temperature coefficient vectors at DHN node i .
$\mathbf{T}_{g,k}^{\text{GS,B}}/\mathbf{T}_{g,k}^{\text{GR,B}}$	Supply/return-temperature coefficient vectors of heat source g .
$\mathbf{T}_{d,k}^{\text{DS,B}}/\mathbf{T}_{d,k}^{\text{DR,B}}$	Supply/return-temperature coefficient vectors of heat load d .
$\boldsymbol{\alpha}_{g,k}^{\zeta,\text{B}}$	Coefficient vector of the convex-combination weight associated with extreme point ζ of CHP unit g .

The centralized IHPS dispatch model is formulated over Bernstein coefficient vectors in each scheduling interval $k \in \mathcal{K}$. The objective function and operating constraints are detailed below.

1) *Objective Function:* The total IHPS operating cost comprises the operating costs of non-CHP and CHP units and the wind-curtailment penalty. The resulting objective function is given by:

$$\min J_{\text{op}}^{\text{B}} = \sum_{i \in \mathcal{I}^{\text{CU}}} J_i^{\text{CU,B}} + \sum_{i \in \mathcal{I}^{\text{wind}}} J_i^{\text{wind,B}} + \sum_{i \in \mathcal{I}^{\text{CHP}}} J_i^{\text{CHP,B}}. \quad (\text{A1})$$

The corresponding individual cost terms are defined as:

$$J_i^{\text{CU},\text{B}} = \sum_{k \in \mathcal{K}} \left[c_{0,i} + \frac{c_{1,i} \mathbf{1}^\top \mathbf{P}_{i,k}^{\text{B}}}{n_t + 1} + \frac{c_{2,i} (\mathbf{1}^\top \mathbf{P}_{i,k}^{\text{B}})^2}{(n_t + 1)^2} \right], \quad \forall i \in \mathcal{I}^{\text{CU}}. \quad (\text{A2})$$

$$J_i^{\text{wind},\text{B}} = \sum_{k \in \mathcal{K}} \sigma_i \left(\bar{\mathbf{P}}_{i,k}^{\text{w},\text{B}} - \frac{\mathbf{1}^\top \mathbf{P}_{i,k}^{\text{w},\text{B}}}{n_t + 1} \right), \quad \forall i \in \mathcal{I}^{\text{wind}}, \quad (\text{A3})$$

$$J_i^{\text{CHP},\text{B}} = \sum_{k \in \mathcal{K}} \left[a_{0,i} + \frac{a_{1,i}}{n_t + 1} \mathbf{1}^\top \mathbf{P}_{i,k}^{\text{B}} + \frac{a_{2,i}}{n_t + 1} \mathbf{1}^\top \mathbf{Q}_{i,k}^{\text{B}} + \frac{a_{3,i}}{(n_t + 1)^2} (\mathbf{1}^\top \mathbf{P}_{i,k}^{\text{B}})^2 + \frac{a_{4,i}}{(n_t + 1)^2} (\mathbf{1}^\top \mathbf{Q}_{i,k}^{\text{B}})^2 \right], \quad \forall i \in \mathcal{I}^{\text{CHP}}. \quad (\text{A4})$$

2) *EPS Operating Constraints*: The EPS operating constraints enforce power balance, transmission security, generation availability, reserve adequacy, and continuous-time ramping limits. For each $k \in \mathcal{K}$, the power-balance equation is expressed as:

$$\sum_{i \in \mathcal{I}^{\text{CU}} \cup \mathcal{I}^{\text{CHP}}} \mathbf{P}_{i,k}^{\text{B}} + \sum_{i \in \mathcal{I}^{\text{wind}}} \mathbf{P}_{i,k}^{\text{w},\text{B}} - \sum_{j \in \mathcal{I}^{\text{WP}}} \mathbf{P}_{j,k}^{\text{PUMP},\text{B}} = \sum_{b \in \mathcal{N}^{\text{bus}}} \mathbf{LD}_{b,k}^{\text{B}}, \quad k \in \mathcal{K}. \quad (\text{A5})$$

The transmission-line flows are constrained by:

$$\left| \frac{1}{n_t + 1} \sum_{b \in \mathcal{N}^{\text{bus}}} S F_{l,b} \left(\sum_{i \in \mathcal{S}_b^{\text{CU}} \cup \mathcal{S}_b^{\text{CHP}}} \mathbf{1}^\top \mathbf{P}_{i,k}^{\text{B}} + \sum_{i \in \mathcal{S}_b^{\text{wind}}} \mathbf{1}^\top \mathbf{P}_{i,k}^{\text{w},\text{B}} - \sum_{j \in \mathcal{S}_b^{\text{WP}}} \mathbf{1}^\top \mathbf{P}_{j,k}^{\text{PUMP},\text{B}} - \mathbf{1}^\top \mathbf{LD}_{b,k}^{\text{B}} \right) \right| \leq F_l, \quad \forall l \in \mathcal{I}^{\text{tl}}, \quad k \in \mathcal{K}. \quad (\text{A6})$$

The generation outputs are bounded by:

$$\underline{P}_i \leq \mathbf{P}_{i,k}^{\text{B}} \leq \bar{P}_i, \quad i \in \mathcal{I}^{\text{CU}} \cup \mathcal{I}^{\text{CHP}}, \quad k \in \mathcal{K}. \quad (\text{A7})$$

The dispatched wind power is bounded by:

$$\mathbf{0} \leq \mathbf{P}_{i,k}^{\text{w},\text{B}} \leq \bar{\mathbf{P}}_{i,k}^{\text{w},\text{B}}, \quad i \in \mathcal{I}^{\text{wind}}, \quad k \in \mathcal{K}. \quad (\text{A8})$$

The upward-reserve variables satisfy the following conditions:

$$\mathbf{0} \leq \mathbf{r} \mathbf{u}_{i,k}^{\text{B}} \leq r_i^{\text{up}} \Delta t, \quad i \in \mathcal{I}^{\text{CU}}, \quad k \in \mathcal{K}, \quad \mathbf{r} \mathbf{u}_{i,k}^{\text{B}} \leq \mathbf{1} \bar{P}_i - \mathbf{P}_{i,k}^{\text{B}}, \quad i \in \mathcal{I}^{\text{CU}}, \quad k \in \mathcal{K}. \quad (\text{A9})$$

Likewise, the downward-reserve variables are subject to:

$$\mathbf{0} \leq \mathbf{r} \mathbf{d}_{i,k}^{\text{B}} \leq r_i^{\text{dn}} \Delta t, \quad i \in \mathcal{I}^{\text{CU}}, \quad k \in \mathcal{K}, \quad \mathbf{r} \mathbf{d}_{i,k}^{\text{B}} \leq \mathbf{P}_{i,k}^{\text{B}} - \mathbf{1} \underline{P}_i, \quad i \in \mathcal{I}^{\text{CU}}, \quad k \in \mathcal{K}. \quad (\text{A10})$$

System-wide reserve adequacy requires:

$$\sum_{i \in \mathcal{I}^{\text{CU}}} \mathbf{1}^\top \mathbf{r} \mathbf{u}_{i,k}^{\text{B}} / (n_t + 1) \geq S R^{\text{up}}, \quad \forall k \in \mathcal{K}, \quad \sum_{i \in \mathcal{I}^{\text{CU}}} \mathbf{1}^\top \mathbf{r} \mathbf{d}_{i,k}^{\text{B}} / (n_t + 1) \geq S R^{\text{dn}}, \quad \forall k \in \mathcal{K}. \quad (\text{A11})$$

Continuous-time ramping is constrained by:

$$-r_i^{\text{dn}} \leq \mathbf{D}_{n_t}^{\text{B}} \mathbf{P}_{i,k}^{\text{B}} \leq r_i^{\text{up}}, \quad i \in \mathcal{I}^{\text{CU}} \cup \mathcal{I}^{\text{CHP}}, \quad k \in \mathcal{K}. \quad (\text{A12})$$

3) *DHN Thermal Constraints*: The algebraic thermal constraints enforce energy conservation at DHN mixing nodes, continuity between nodal and pipeline inlet temperatures, and heat balances at the sources and loads. The supply- and return-network temperature-mixing equations take the following forms:

$$\sum_{l \in \mathcal{S}_i^{\text{PS}-}} (\mathbf{T}_{l,k}^{\text{OS},\text{B}} \mathbf{m}_{l,k}^{\text{S},\text{B}}) + \sum_{g \in \mathcal{S}_i^{\text{HS}}} (\mathbf{T}_{g,k}^{\text{GS},\text{B}} \mathbf{m}_{g,k}^{\text{G},\text{B}}) = \mathbf{T}_{i,k}^{\text{NS},\text{B}} \left(\sum_{l \in \mathcal{S}_i^{\text{PS}-}} \mathbf{m}_{l,k}^{\text{S},\text{B}} + \sum_{g \in \mathcal{S}_i^{\text{HS}}} \mathbf{m}_{g,k}^{\text{G},\text{B}} \right), \quad \forall i \in \mathcal{N}^{\text{nd}}, \quad k \in \mathcal{K}. \quad (\text{A13})$$

$$\sum_{l \in \mathcal{S}_i^{\text{PR}-}} (\mathbf{T}_{l,k}^{\text{OR},\text{B}} \mathbf{m}_{l,k}^{\text{R},\text{B}}) + \sum_{d \in \mathcal{S}_i^{\text{HL}}} (\mathbf{T}_{d,k}^{\text{DR},\text{B}} \mathbf{m}_{d,k}^{\text{D},\text{B}}) = \mathbf{T}_{i,k}^{\text{NR},\text{B}} \left(\sum_{l \in \mathcal{S}_i^{\text{PR}-}} \mathbf{m}_{l,k}^{\text{R},\text{B}} + \sum_{d \in \mathcal{S}_i^{\text{HL}}} \mathbf{m}_{d,k}^{\text{D},\text{B}} \right), \quad \forall i \in \mathcal{N}^{\text{nd}}, \quad k \in \mathcal{K}. \quad (\text{A14})$$

The inlet temperature of each outgoing pipeline is linked to its associated nodal temperature through:

$$\mathbf{T}_{l,k}^{\text{IS},\text{B}} = \mathbf{T}_{i,k}^{\text{NS},\text{B}}, \quad i \in \mathcal{N}^{\text{nd}}, \quad l \in \mathcal{S}_i^{\text{PS}+}, \quad k \in \mathcal{K}, \quad (\text{A15})$$

$$\mathbf{T}_{l,k}^{\text{IR},\text{B}} = \mathbf{T}_{i,k}^{\text{NR},\text{B}}, \quad i \in \mathcal{N}^{\text{nd}}, \quad l \in \mathcal{S}_i^{\text{PR}+}, \quad k \in \mathcal{K}. \quad (\text{A16})$$

The source and load heat balances are expressed by:

$$\mathbf{Q}_{g,k}^{\text{B}} = c \mathbf{m}_{g,k}^{\text{G},\text{B}} (\mathbf{T}_{g,k}^{\text{GS},\text{B}} - \mathbf{T}_{g,k}^{\text{GR},\text{B}}), \quad g \in \mathcal{S}_i^{\text{HS}}, \quad k \in \mathcal{K}. \quad (\text{A17})$$

$$\mathbf{Q}_{d,k}^{\text{B}} = c \mathbf{m}_{d,k}^{\text{D},\text{B}} (\mathbf{T}_{d,k}^{\text{DS},\text{B}} - \mathbf{T}_{d,k}^{\text{DR},\text{B}}), \quad d \in \mathcal{S}_i^{\text{HL}}, \quad k \in \mathcal{K}. \quad (\text{A18})$$

The continuous pipeline temperature fields are further governed by the heat-transport PDE constraints presented in the main paper.

4) *DHN Hydraulic Constraints*: The hydraulic constraints enforce mass conservation at each DHN node and the pressure-loss relations along the pipelines. The corresponding mass-flow balances in the supply and return networks can be written as:

$$\sum_{l \in \mathcal{S}_i^{\text{PS}-}} \mathbf{m}_{l,k}^{\text{S},\text{B}} + \sum_{g \in \mathcal{S}_i^{\text{HS}}} \mathbf{m}_{g,k}^{\text{G},\text{B}} = \sum_{l \in \mathcal{S}_i^{\text{PS}+}} \mathbf{m}_{l,k}^{\text{S},\text{B}} + \sum_{d \in \mathcal{S}_i^{\text{HL}}} \mathbf{m}_{d,k}^{\text{D},\text{B}}, \quad \sum_{l \in \mathcal{S}_i^{\text{PR}-}} \mathbf{m}_{l,k}^{\text{R},\text{B}} + \sum_{d \in \mathcal{S}_i^{\text{HL}}} \mathbf{m}_{d,k}^{\text{D},\text{B}} = \sum_{l \in \mathcal{S}_i^{\text{PR}+}} \mathbf{m}_{l,k}^{\text{R},\text{B}} + \sum_{g \in \mathcal{S}_i^{\text{HS}}} \mathbf{m}_{g,k}^{\text{G},\text{B}}, \quad \forall i \in \mathcal{N}^{\text{nd}}, \quad k \in \mathcal{K}. \quad (\text{A19})$$

The supply and return network pressure head drops obey the following Darcy–Weisbach equations:

$$\mathbf{pr}_{i_1,k}^{\text{S},\text{B}} - \mathbf{pr}_{i_2,k}^{\text{S},\text{B}} = \mu_l (\mathbf{m}_{l,k}^{\text{S},\text{B}})^2, \quad \mathbf{pr}_{i_1,k}^{\text{R},\text{B}} - \mathbf{pr}_{i_2,k}^{\text{R},\text{B}} = \mu_l (\mathbf{m}_{l,k}^{\text{R},\text{B}})^2, \quad \forall i_1 = \text{Nd}_l^{\text{in}}, \quad i_2 = \text{Nd}_l^{\text{out}}, \quad l \in \mathcal{I}^{\text{DP}}, \quad k \in \mathcal{K}. \quad (\text{A20})$$

5) *CHP and Pump Coupling Constraints:* The CHP outputs and the associated convex-combination weights satisfy:

$$\begin{aligned}
 \mathbf{P}_{g,k}^B &= \sum_{\zeta=1}^{NK_g} P_g^\zeta \boldsymbol{\alpha}_{g,k}^{\zeta,B}, \quad \mathbf{Q}_{g,k}^B = \sum_{\zeta=1}^{NK_g} Q_g^\zeta \boldsymbol{\alpha}_{g,k}^{\zeta,B}, \\
 \forall g \in \mathcal{I}^{\text{CHP}}, \quad k \in \mathcal{K}, \quad \sum_{\zeta=1}^{NK_g} \boldsymbol{\alpha}_{g,k}^{\zeta,B} &= \mathbf{1}, \\
 \mathbf{0} \leq \boldsymbol{\alpha}_{g,k}^{\zeta,B} \leq \mathbf{1}, \quad \zeta &= 1, \dots, NK_g.
 \end{aligned} \tag{A21}$$

The electrical power consumed by water pump j is determined from the corresponding mass flow and the pressure-head difference between the supply and return networks:

$$\begin{aligned}
 \mathbf{P}_{j,k}^{\text{PUMP},B} &= \frac{1}{\rho \eta_j} \mathbf{m}_{j,k}^{\text{G},B} \left(\mathbf{pr}_{i,k}^{\text{S},B} - \mathbf{pr}_{i,k}^{\text{R},B} \right), \\
 j \in \mathcal{I}^{\text{WP}}, \quad i &= \text{Nd}_j^{\text{WP}}, \quad k \in \mathcal{K}.
 \end{aligned} \tag{A22}$$

Equations (A5)–(A22), together with the pipeline heat-transport PDE constraints, form the detailed finite-dimensional centralized IHPS dispatch model.